

SUMMARY

Senior Computer Vision & AI engineer (7 years) specializing in **video understanding at scale** — detection, multi-object tracking, re-identification, and action recognition across live surveillance, sports, broadcast, and microscopy video — with production experience in real-time streaming pipelines and edge deployment. Technical lead who has shipped research to production and led teams of up to 8.

EDUCATION

Nile University M.S. in Communication and Information Technology, GPA: 4.0/4.0	Giza, Egypt 2021–2026
Minia University B.S. in Biomedical Engineering, Grade: Very Good with Honors (84.6%)	Minya, Egypt 2014–2019

EXPERIENCE

CyShield Senior Computer Vision Engineer (<i>Full-time</i>)	New Cairo, Egypt January 2025 – Current
– Stadium Video Surveillance: Delivered a multi-camera stadium surveillance system on live video streams — person detection, multi-object tracking and re-identification across overlapping camera views, and action recognition for crowd-incident and prohibited-behaviour alerting. – Biometric Border Control: Built face- and fingerprint-based identity verification for border checkpoints — face detection with quality/liveness filtering on camera feeds, embedding extraction and fingerprint minutiae matching, and 1:1 verification plus 1:N watchlist identification under strict latency and false-match limits. – Generative & Vision-Language Models: Trained diffusion- and GAN-based inpainting models for image restoration and content-aware editing (LPIPS, FID, SSIM, PSNR evaluation), and fine-tuned VLMs for grounding, captioning, and visual QA on in-house data. – Technical Leadership: Led a squad of 3 AI engineers — scoping, code review, mentoring, and aligning research direction with product deadlines.	
CNTXT AI/Computer Vision Consultant (<i>Part-time</i>)	Remote — UAE / Egypt January 2026 – Current
– Video Data & QA at Scale: Built an egocentric data-collection and QA platform processing 100+ hours of 3-camera teleoperation video per week into validated LeRobot datasets — 3D hand-pose reconstruction, automated video QA, and VLM-based action annotation. – Multi-modal CV & Speech: Built instance-segmentation systems and cross-modal RGB/IR drone-image alignment; designed a multi-stage Arabic ASR pipeline (denoising, diarization, dialect-aware transcription). – Agentic RAG (Arabic): Shipped a LangGraph retrieval service — evaluator–optimizer loop, hybrid dense/BM25 search, cross-encoder reranking, grounded citations; Hit@1 0.84 → 0.95.	
Lumin Soft AI Team Lead (<i>Full-time</i>)	Giza, Egypt November 2023 – January 2025
– Sports Video Analytics (flagship): Led a large-scale sports video-analysis system — player and ball detection & tracking, team clustering, re-identification, action recognition, and possession tracking across full-match footage — producing annotated video and a per-match metrics report. – Team Leadership: Led a team of 8 AI developers across segmentation, detection, tracking, and OCR projects — running 1:1s, sprint planning, and technical mentorship. – Research-to-Production: Delivered 3 production video and imaging projects (segmentation, detection, tracking, clustering, ReID, OCR), turning research prototypes into deadline-bound production pipelines.	

Applied Innovation Center

Giza, Egypt

R&D Engineer (*Full-time*)

October 2022 – November 2023

- Segmented land cover into 7 classes from Sentinel-2 and Planet satellite imagery with U-Net and Transformer networks (98% overall accuracy), on end-to-end geospatial pipelines — preprocessing, tiling, inference, post-processing.

Center for Informatics Science, Nile University

Giza, Egypt

Research Assistant (*Full-time*)

April 2021 – April 2023

- Classified rice crop diseases across 5 classes with ResNet and Transformer models, achieving 96% F1 on a 250K-image dataset; competed in CV challenges (UW-Madison GI-tract segmentation).

Dawarha

Cairo, Egypt

Computer Vision Engineer (*Part-time*)

April 2021 – September 2023

- Built a real-time YOLOv3 + KMeans video pipeline for 2-class recyclable-material detection (90 mAP), [deployed on a Raspberry Pi at 11 FPS](#).

VLens

Cairo, Egypt

Computer Vision Engineer (*Part-time*)

February 2021 – November 2021

- Built national-ID OCR with U-Net text segmentation and KMeans preprocessing, [deployed on AWS EC2](#).

Affectiva

Massachusetts, USA

Mentor (*Contract, 3 months*)

June 2020 – August 2020

- Mentored student teams through CV project scoping, model selection, and demo delivery.

TechnoIsland

Cairo, Egypt

Computer Vision Engineer (*Part-time*)

October 2019 – November 2021

- Built a video pipeline for sperm-sample analysis — YOLOv3 detection, Yolact instance segmentation, and DeepSort tracking across microscopy video frames.

PROJECTS

TV Ad-Recognition Pipeline

Real-time broadcast video recognition via multi-modal fingerprinting

- **Video & audio fingerprinting:** Matched live TV broadcasts against an ad catalog using scene-change keyframes (PySceneDetect + pHash + CLIP ViT-B/32 embeddings), audio fingerprinting (Chromaprint), and speaker embeddings (WavLM x-vectors), fused over Elasticsearch kNN + BM25 with NMS-suppressed airing events.
- **Live streaming:** Added a persistent-socket WebSocket endpoint for continuous per-channel recognition. *Stack:* two FastAPI services, Elasticsearch, PyTorch, FFmpeg; deployed as auto-start Windows services.

Sperm Analysis (CASA)

Computer-aided semen analysis from microscopy video

- **Video analysis engine:** Built a CASA pipeline measuring sperm count, motility (per-cell kinematics via multi-object tracking), and morphology from microscopy and industrial-camera video — YOLO detection served through ONNX Runtime.
- **Product & deployment:** Shipped as an offline Windows desktop app — FastAPI backend (PyInstaller) in an Electron + React UI, industrial-camera capture (GenTL / Harvesters), encrypted model weights, HTTPS licensing service, and PDF clinical reports.

BlacksmithAI (*Freelance, 2026*)

blacksmithai.org

Natural-language to CAD agentic pipeline

- Designed a multi-stage agent state machine (decompose → plan → assemble → execute → validate → repair) over a deterministic JSON IR with content-addressed caching, multi-modal image grounding, and human-in-the-loop review; Pydantic-validated tool catalog with pure verifier tools for LLM self-checking. *Stack:* FastAPI, Qdrant + BGE RAG, CadQuery, React + react-three-fiber, Docker.

RESEARCH

Fatima Fellowship

Research Intern (*Part-time*)

Stanford, USA

September 2023 – August 2024

- Developed generative models to enhance low-resolution MRI images, focusing on model optimization and validation for clinical applications.

Linköping University

Research Intern (*Part-time*)

Linköping, Sweden

October 2022 – January 2023

- Developed GANs to manipulate Maxar aerial images by adding realistic seasonal effects, collaborating with geospatial analysts to ensure accuracy.

PUBLICATIONS

- [1] T. A. Abdel Salam, A. M. Abdelsalam, N. R. Sholkamy, and **Sebaq, Ahmad**, “Assessment of artificial intelligence in detecting errors on panoramic radiographs”, *Imaging Science in Dentistry*, vol. 56, e12, 2026. DOI: [10.5624/isd.20250242](https://doi.org/10.5624/isd.20250242)
- [2] **Sebaq, Ahmad** and M. ElHelw, “Rsdiff: Remote sensing image generation from text using diffusion model”, *Neural Computing and Applications*, vol. 36, no. 36, pp. 23 103–23 111, 2024. DOI: [10.1007/s00521-024-10363-3](https://doi.org/10.1007/s00521-024-10363-3)
- [3] Y. A. Alnaggar, **Sebaq, Ahmad**, K. Amer, E. Naeem, and M. Elhelw, “Rice plant disease detection and diagnosis using deep convolutional neural networks and multispectral imaging”, in *International Conference on Model and Data Engineering*, Springer, 2023, pp. 16–25. DOI: [10.1007/978-3-031-21595-7_2](https://doi.org/10.1007/978-3-031-21595-7_2)
- [4] M. S. M. Bahgat and **Sebaq, A. M.**, “An Analytical Computational Algorithm for Solving a System of Multipantograph DDEs Using Laplace Variational Iteration Algorithm”, in *Advances in Astronomy*, Jun. 2021, p. 16. DOI: [10.1155/2021/7741166](https://doi.org/10.1155/2021/7741166)

AWARDS

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| • Full Master Scholarship from Nile University | 2021–2023 |
| • 3rd Place, Global Monitoring for Environment and Security (GMES) & Africa Incubation Challenge | 2021 |
| • 3rd Place in Affectiva Makeathon with Inside Out project | 2020 |
| • Nokia Africathon Finalist with ALS project | 2019 |

SKILLS

- **Video Understanding:** Detection & multi-object tracking (YOLO, Yolact, DeepSort), cross-camera re-identification, action recognition, scene-change detection (PySceneDetect), keyframe / perceptual hashing (pHash), CLIP video-frame embeddings, large-scale automated video QA
- **Computer Vision:** OpenCV; segmentation (U-Net, Mask R-CNN, Swin), face & fingerprint biometrics (1:1 verification, 1:N identification), OCR, 3D hand-pose / SLAM, cross-modal RGB/IR alignment
- **Real-time & Video Infrastructure:** FFmpeg, live stream ingest, WebSocket / SSE streaming services, industrial-camera capture (GenTL / Harvesters), ONNX Runtime, edge deployment (Jetson Nano & Orin, Raspberry Pi)
- **ML / Deep Learning:** PyTorch, HuggingFace, scikit-learn; CNNs, Transformers, diffusion models, GANs; audio fingerprinting, ASR (Whisper), speaker diarization (Pyannote), denoising (DeepFilterNet3)
- **Generative AI & LLMs:** LLM orchestration (LangGraph), agentic RAG, multi-modal VLMs, vector & hybrid search (Qdrant, Elasticsearch), tool-use / function-calling
- **Programming & MLOps:** Python, C++, JavaScript / TypeScript, React; FastAPI, Docker, Jenkins CI, AWS (EC2), Git; desktop packaging (Electron, PyInstaller); QGIS, Sentinel-2 / Planet / Maxar imagery